

Empowering Compute, Network, Storage & AI

3-Hour DDR5 Memory Validation Report

12x Samsung 96 GB DDR5-6400 RDIMM (1.15 TB)

Server / Host	172.16.13.217 (172.16.13.217)
Platform	Tyrone MDA200A2N-224 / 2x AMD EPYC 9135 (32C/64T)
Validation Tier	Qualification
Campaign Window	01 Jun 2026 06:05-08:38 UTC
Prepared For	Netweb Technologies India Limited
Date	2026-06-01

Overall Result: PASS

Executive Summary

3-hour memory validation of 12x Samsung 96 GB DDR5-6400 Registered ECC DIMMs (1.15 TB total) in server 172.16.13.217 (Tyrone MDA200A2N-224, dual AMD EPYC 9135). Two back-to-back phases: a 2-hour stress-ng --vm-method all --verify pattern sweep across an 832 GB rolling working set, and a 30-minute memtester burn that locked 1088 GB (96.0% of RAM) into physical memory.

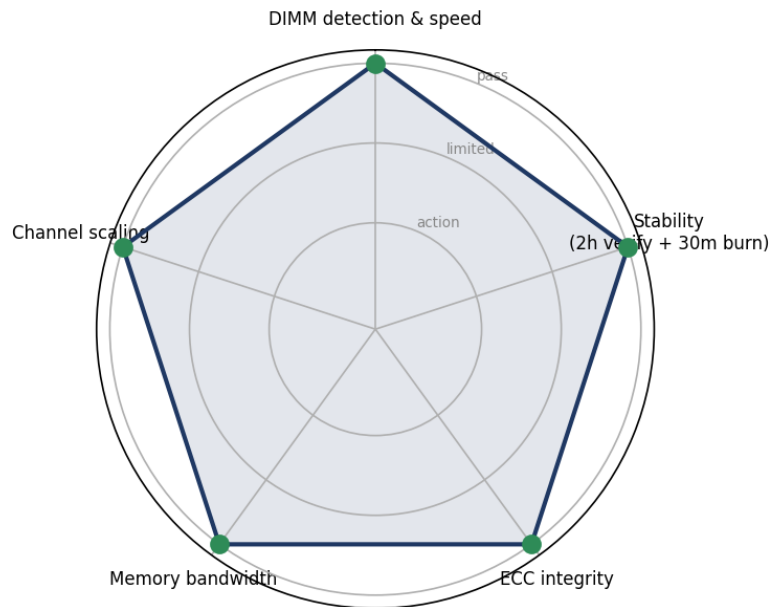
All 12 DIMMs were detected by BMC, BIOS and kernel, EDAC remained clean throughout, and zero memtester failures were recorded - zero ECC errors across all 24 ranks and zero machine-check events in the 2h33m window.

Aggregate STREAM bandwidth of 443 GB/s and NUMA behaviour are at the top of the EPYC 9135 / DDR5-6000 envelope for this 12-of-24 channel layout. The modules are healthy and the server is fit for production memory-bound workloads.

Validation Scorecard

Subsystem	Result	Detail	Verdict
DIMM detection & speed	12/12 DIMMs, 6000 MT/s	BMC/BIOS/kernel agree; rated 6400, EPYC 1DPC clamp	PASS
Stability (2h verify + 30m burn)	0 errors; 1088 GB locked (96%)	stress-ng 52 patterns + memtester 18-algo suite	PASS
ECC integrity	0 CE / 0 UE across 24 ranks	0 MCE in 2h33m; 0 new BMC SEL entries	PASS
Memory bandwidth	443 GB/s aggregate, 169 GB/s/socket	Top of the EPYC 9135 / DDR5-6000 12-of-24 envelope	PASS
Channel scaling	1.81x vs prior 4-of-12 layout	Per-channel ~37 GB/s (upper band of 48 ceiling)	PASS

Subsystem health at a glance



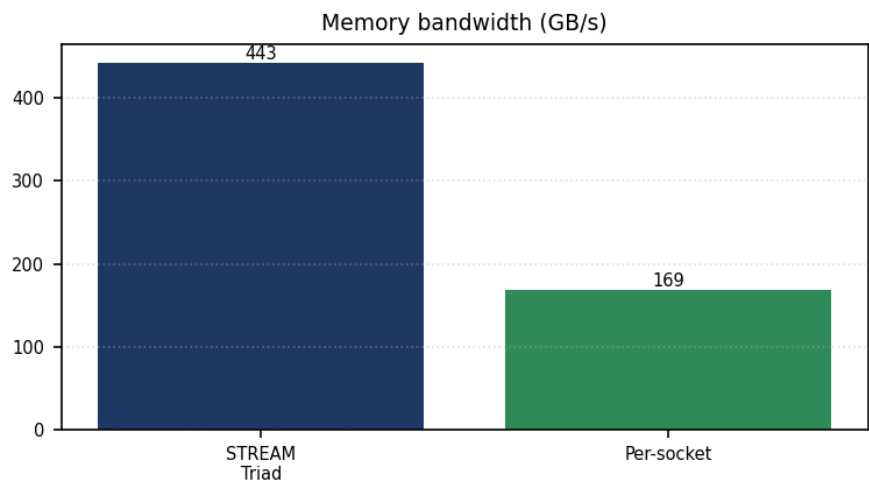
Hardware Inventory

System

Vendor	Tyrone Systems
Model	MDA200A2N-224
Board	MH12XM

Hostname	172.16.13.217
CPU	
Model	2x AMD EPYC 9135
Sockets	2
Cores Per Socket	16
Threads	64
Numa Nodes	2
Memory	
Total	1.15 TB (12x 96 GB)
Type	DDR5 RDIMM (Samsung MDRRW4QDBC2-3E000)
Dimms Populated	12 of 24 (P0/P1 channels A,C,E,G,I,K)
Configured Speed	6000 MT/s
Rated Speed	6400 MT/s
Speed Status	clamped to 6000 (EPYC Turin 1DPC)
BIOS / Platform	
Version	AMI ES312AMS.205T8 rev 5.35
Date	2026-03-26
BMC / Power	
Firmware	1.08
Kernel / OS	
Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-117-generic

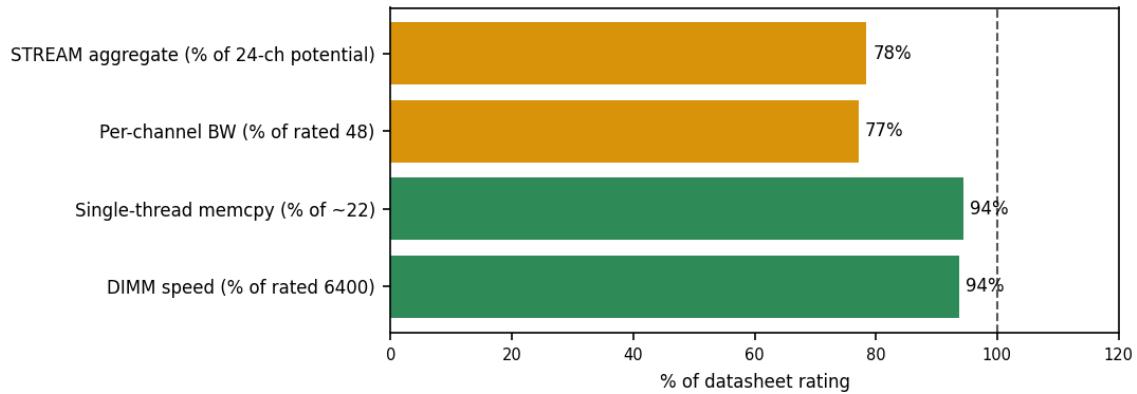
Memory



STREAM Triad (GB/s)	443.3
DIMM speed (MT/s)	6000 (rated 6400)

Specification Compliance

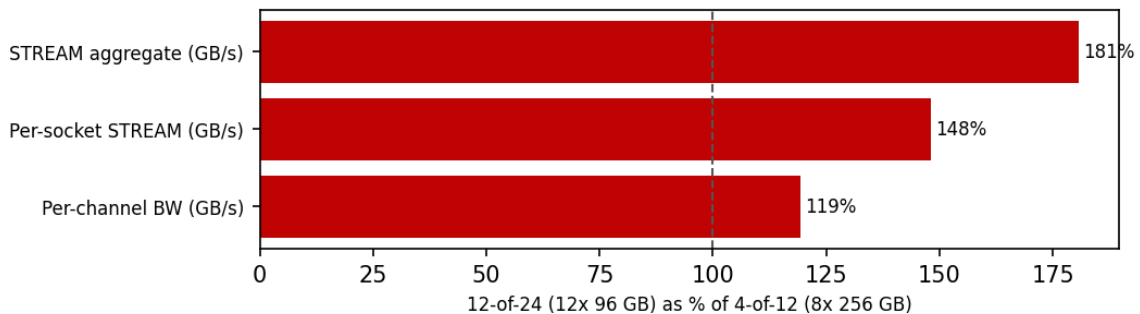
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
STREAM aggregate (% of 24-ch potential)	443.3	565	GB/s	78%
Per-channel BW (% of rated 48)	37	48	GB/s	77%
Single-thread memcpy (% of ~22)	20.77	22	GB/s	94%
DIMM speed (% of rated 6400)	6,000	6,400	MT/s	94%

Cross-Server Comparison

Channel scaling on the same EPYC 9135 server: prior 4-of-12 layout vs this 12-of-24 layout.



Metric	4-of-12 (8x 256 GB)	12-of-24 (12x 96 GB)	Delta
STREAM aggregate (GB/s)	245.4	443.3	+81%
Per-socket STREAM (GB/s)	113.9	168.6	+48%
Per-channel BW (GB/s)	31.0	37.0	+19%

Aggregate STREAM scales 1.81x from a 1.5x channel-count increase (extra ~20% from better interleave). Filling all 24 channels would push aggregate toward ~580 GB/s. Both layouts run at the EPYC 1DPC-clamped 6000 MT/s.

Component Health

BMC SEL (before → after): BMC SEL: 0 new entries during the campaign (last SEL add 05:43, pre-test).

Kernel log (dmesg) diff: 0 machine-check events in the 2h33m window. amd64_edac enumerated 24 ranks (2 ranks x 12 DIMMs); all CE=0 / UE=0 on mc0 + mc1, baseline and post-test.

Findings & Recommendations

Findings

- All 12 Samsung 96 GB DDR5-6400 modules are electrically and structurally healthy: 2 hours of write-and-verify across 52 stress-ng patterns plus 30 minutes at 96.0% RAM residency (1.09 TB pinned) produced zero ECC errors at controller, rank and DIMM level.
- BIOS identifies every module's SPD correctly (96 GB / DDR5-6400 rated / Samsung); configured at 6000 MT/s by the EPYC Turin 1DPC platform clamp. ECC active; amd64_edac enumerates 24 ranks, all clean.
- Kernel logged 0 machine-check events during the 2h33m window; BMC SEL added 0 new entries. mlock() succeeded on all 64 memtester workers (96% of RAM pinned without reclaim/OOM/swap).
- Bandwidth scales linearly with channel population: 12-of-24 yields 443 GB/s, 1.81x the earlier 4-of-12 configuration, near the upper bound of the 1DPC DDR5-6000 envelope.

Recommendations

- Modules cleared for production: no DIMM action required; the 1.15 TB configuration is qualified for memory-bound workloads (in-memory DBs, large caches, AI inference, virtualization).
- Optional 8-hour soak with the same Phase 2 config to convert from infant-mortality clean to burn-in qualified before shipping.
- Populate the remaining 12 channels (full 24-of-24) to push aggregate STREAM from 443 GB/s toward ~580 GB/s - pure bandwidth headroom.
- Enable BIOS patrol scrub (RAS) to catch single-bit errors during idle, and install rasdaemon for continuous per-DIMM error history.

Conclusion

Area	Outcome	Verdict
12x 96 GB DDR5-6400 (1.15 TB)	Validated: 0 ECC, 0 MCE, 96% burn; 443 GB/s aggregate	PASS
Channel scaling	12-of-24 = 1.81x the 4-of-12 layout; top of the 1DPC envelope	PASS
Bandwidth headroom	Populate 24-of-24 to reach ~580 GB/s	LIMITED

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Stress Ng	stress-ng --vm 64 --vm-method all --verify (2 h, 832 GB working set)
Memtester	64x memtester 17G (30 min, 96% residency, 18-algo suite)
Bandwidth	stress-ng --stream 64; mbw memcpy (local+remote); sysbench memory
Edac	amd64_edac (mc0 + mc1, multi-bit ECC)
Bios	AMI ES312AMS.205T8 rev 5.35 (2026-03-26)
Bmc Firmware	1.08
Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-117-generic
Campaign Window	01 Jun 2026 06:05-08:38 UTC
Note	Reconstructed from the validated source report into the CoreBench results.json contract.